Can Ambient WiFi Replace LiDAR for Automotive SLAM? Localization Yes, Mapping No — and Why

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Abstract—**LiDAR dominates the sensor cost of an autonomous vehicle, which motivates a blunt question: can ambient WiFi sensing simply take its place? We answer it with the first head-to-head, on-vehicle, outdoor comparison of WiFi and LiDAR SLAM evaluated on the *same* ray-traced scenes, against the *same* ground truth, with the *same* six metrics. The study is simulation-based (Sionna RT), anchored by a real-LiDAR control: our SLAM back-end run on KITTI odometry attains 1.16 m aligned ATE over a 394 m drive ($\approx 0.3\%$ drift), which is consistent with real LiDAR odometry. Three findings follow. First, on *localization* commodity-CSI WiFi reaches *parity* with LiDAR — 0.098 m ATE against 0.102 m for our geometric LiDAR model in sparse multipath, and decimetre-class accuracy in general — at 84–600× lower sensor cost. Second, on *mapping* it fails outright (occupancy IoU ≈ 0 versus up to 1.00 for LiDAR), and no price buys coverage it cannot produce. Third, we locate the mapping ceiling precisely: it is *not* path discrimination, as previously supposed, but a front-end limit — roughly 89% of MUSIC detections correspond to *no real propagation path*, and a 6.45 m median range *bias* (far beyond the 1.87 m resolution limit) corrupts the rest. Consequently a learned discriminator cannot repair the map, and we show that a heuristic, a random forest, and a neural network all fail identically. The practical consequence is a hybrid: adding WiFi to an existing LiDAR costs $\approx 0.5\%$ more and improves localization by 36–79% — provided the two sensors are of comparable accuracy, since a badly mismatched pair degrades the stronger one.**

Index Terms—**Internet of Things, SLAM, Wi-Fi, channel state information, LiDAR, sensor fusion, autonomous vehicles, cost analysis, ray tracing.**

I. Introduction

LiDAR is the reference exteroceptive sensor for vehicle SLAM, and it is also the expensive one: a mid-range spinning unit of the class routinely used in research costs thousands of dollars, while the WiFi silicon already present in a vehicle costs tens. If ambient WiFi — signals that are radiated anyway, by infrastructure that already exists — could stand in for LiDAR, the sensing bill for autonomy would change by orders of magnitude. That is the question this paper asks, and it asks it without hedging: *can WiFi replace LiDAR, or not?*

The honest answer turns out to be neither "yes" nor "no" but a boundary, and the boundary is the contribution. WiFi replaces LiDAR for one half of SLAM and fails at the other, and the failure is not where the literature — including our own earlier work [1] — assumed it was.

A. Contributions

1) **The first on-vehicle, outdoor, head-to-head WiFi-vs-LiDAR SLAM comparison.** Both modalities are evaluated on identical ray-traced scenes against identical footprint ground truth with an identical metric suite, and the LiDAR back-end is externally validated on real KITTI data [2] ($\approx 0.3\%$ drift). Prior WiFi-SLAM is indoor [3], or uses WiFi to *assist* a primary ranging sensor [4], [5]; none tests replacement on a vehicle (Section II).
2) **A sensor-cost analysis** — to our knowledge the first for WiFi-vs-LiDAR SLAM — including *cost-normalized* accuracy and coverage. WiFi's sensing package is **84–600×** cheaper than the LiDAR class we simulate, and delivers one to two orders of magnitude better accuracy per dollar (Section VI).
3) **A symmetric fusion study with a cost verdict.** Fusing the two helps *conditionally*, and we identify the condition (sensor parity): fusion beats both solo sensors in three of four configurations but *degrades* the stronger sensor by 8× when the pair is badly mismatched (Section VII).
4) **The mapping ceiling, located and quantified.** That WiFi ranges poorly [6] and that RF sensors produce spurious returns [7] are both known qualitatively. We contribute the *first quantification*: an isolation experiment attributes $\approx 89\%$ of detections to *phantoms* (no real propagation path) and shows the residual error is a 6.45 m **bias, not a resolution limit** — so bandwidth does not fix it. Path discrimination, the previously assumed culprit, is the *smallest* of the three terms. This refines the interpretation of [1] and tells a learned method where it must act (Section VIII).

B. Relationship to our prior work

This paper extends [1], which established ambient WiFi as a feasible *radar*-replacement sensing front-end and reported an oracle-sensing map at 0.25 m accuracy alongside a realistic-sensing map bounded near 5 m.

This paper is nonetheless *self-contained*: every WiFi result reported here is regenerated by our own runs (Table I, quoted as mean ± std over five seeds) rather than imported, so no

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Code, data, and every figure-generating script are released openly; all reported numbers are reproduced from the committed artifacts.

claim depends on a result the reader cannot verify from the artifacts released with *this* paper.

Section VIII refines one of the earlier work's *interpretations*: it attributed the realistic-mapping floor to path discrimination and argued the discrimination was learnable. The isolation experiment below shows discrimination is in fact the *smallest* of three mechanisms, and that a learned discriminator therefore cannot lift the map.

II. Related Work

WiFi as a standalone SLAM sensor. P2SLAM [3] is the closest prior art: it extracts two-way bearings from commodity CSI and integrates them with odometry in a GraphSLAM back-end, achieving 26.9 cm median tracking. It is, however, evaluated *indoors only* (25×30 m and 50×40 m, with fixed anchors) and is benchmarked against *visual* SLAM, not LiDAR. Radio-fingerprint SLAM [8] does run outdoors on a mobile platform and uses radio as the primary modality, but it relies on RSS fingerprinting from smartphones rather than CSI, runs on a slow research UGV rather than a road vehicle, and attains its best accuracy only when fused with LiDAR.

WiFi as an assistant to a primary sensor. A larger body of work uses WiFi to *augment* a ranging sensor rather than replace it. ViWiD [4] offloads loop closure from a visual back-end to WiFi; [5] supplies WiFi fingerprint loop closures to a LiDAR SLAM that still performs the mapping; [9] fuses WiFi RSSI, LiDAR and IMU in an EKF. In every case LiDAR or a camera remains the mapping sensor. None of these works reports a WiFi-vs-LiDAR SLAM accuracy comparison, and none treats WiFi as a candidate replacement.

Learned RF sensing. Deep models can recover geometry from RF: RF-Pose [10] estimates human pose through walls from WiFi-band reflections, [11] regresses dense body pose from commodity CSI, and [12] reconstructs 3-D point clouds from CSI with a transformer. Notably these methods learn *end-to-end from raw CSI*; none classifies estimated propagation paths. Section VIII explains why that distinction is decisive.

What is already known about WiFi's range. The passive-WiFi-radar community has long held that WiFi's channel bandwidth makes *range* unusable: Li et al. [6] report that the 20/40 MHz bandwidths give 50/25 ns time resolution and hence "distance estimation errors in the order of several meters", and conclude that "due to the limited bandwidth in WiFi, the range resolution is not sufficient" — so they abandon range and use Doppler alone. Our pipeline is deliberately range-based, and Section VIII measures precisely what that costs. Likewise, spurious returns are familiar in radar: RadarSLAM [7] notes that naive peak detection yields peaks "distributed randomly across the whole radar image, even for the areas with no real object". Neither literature, however, *quantifies* the phenomenon or separates it from the other failure modes — which is what Section VIII does.

The open cell. On-vehicle, outdoor, commodity-CSI WiFi sensing evaluated head-to-head against LiDAR for SLAM — with a cost analysis — is, to our knowledge, unoccupied. This paper occupies it.

III. System Model

Both modalities are driven through one pipeline and scored on one plane, so that any difference between them is attributable to the sensor and not to the estimator, the scene, or the metric (Fig. 1).

Scenes and ground truth. Two ray-traced outdoor scenes are used throughout: a *controlled wall* (a large metal reflector, giving clean single-bounce geometry) and a reflective *street canyon* with buildings and cars. Ground truth is the bird's-eye footprint of every non-floor scatterer, identical for both sensors.

WiFi path. Ambient access points illuminate the scene; the vehicle receives commodity CSI, from which a joint 2-D (delay–azimuth) MUSIC front-end estimates each path's bistatic delay and arrival azimuth. Each detection is triangulated on the bistatic ellipse — the reflector lies where the measured path length $|\mathrm{AP} \to R| + |R \to \mathrm{veh}|$ and the arrival bearing intersect — and the resulting reflectors, together with the same detections used as particle weights, drive a Rao-Blackwellized particle-filter SLAM.

LiDAR path. The LiDAR is reduced to the same plane: a horizontal ring produces a 2-D scan, which a scan-to-map ICP registers against an accumulated voxel map. Because a single LiDAR model would beg the question of *which* LiDAR, we bracket the sensor with two models (Section IV).

The common comparison plane. Trajectories are xy, maps are 2-D point sets, and both sensors are scored with the same six metrics: ATE and RPE for the trajectory; Chamfer distance, directional map-accuracy (estimate$\to$truth) and map-completeness (truth$\to$estimate), and occupancy IoU for the map. Reducing LiDAR to 2-D is a deliberate, stated choice: it is what makes the comparison apples-to-apples, and it is discussed as a limitation in Section IX.

IV. LiDAR Baselines and Their External Validation

A comparison against "LiDAR" is only as honest as the LiDAR one chooses. We therefore bracket the sensor with two models rather than privileging one.

Model A (geometric). Each scene object is sliced at the scan plane into wall segments and ray-cast with occlusion-correct nearest-hit. This yields crisp, precise returns — and, because it sees only what a horizontal ring can see, sparse coverage.

Model B (physical). The LiDAR is modelled as a monostatic node co-located with the vehicle, with scene materials given a diffuse scattering coefficient; the single-scatter interaction vertices of the ray tracer are the returns. This yields dense coverage — and noisier points.

Both run at the parameters of a mid-range spinning unit (120 m range, ± 3 cm, 360°). The two models bracket reality: A is optimistic on precision and pessimistic on coverage, B the reverse, and a real LiDAR sits between them. Reporting a single baseline would have concealed this, so LiDAR is reported throughout as an *envelope*.

External validation. A simulated LiDAR is only persuasive if the back-end that consumes it behaves like a real one.

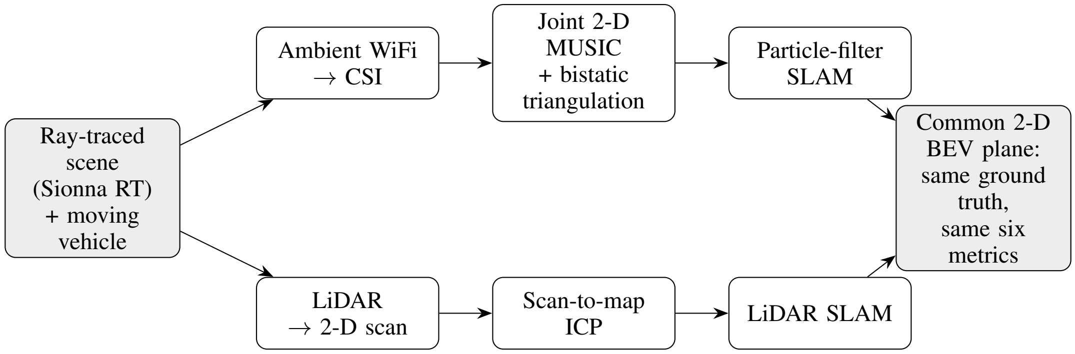

Fig. 1. Both sensors are driven through the same scene and scored on the same 2-D bird's-eye plane, against the same footprint ground truth, with the same six metrics. Any difference is then attributable to the sensor, not to the estimator or the metric.

We therefore ran the *same* scan-to-map ICP on real KITTI odometry [2] (sequence 04; 271 frames; 394 m): it attains 0.154 m RPE per frame and 1.16 m aligned ATE, i.e. $\approx 0.3\%$ translational drift — squarely in the range reported for real LiDAR odometry. The back-end producing our LiDAR rows is therefore sound, and the LiDAR results below are not an artefact of the simulator.

V. Accuracy: WiFi vs. LiDAR

Table I and Fig. 2 report both sensors on both scenes. WiFi rows are this paper's own runs, quoted as mean $\pm$ standard deviation over five seeds rather than a single realization, because the WiFi pipeline is stochastic and a single run is not a claim. Two facts dominate, and they point in opposite directions.

On localization WiFi reaches parity with LiDAR — for about one percent of the price. In the sparse-multipath controlled scene, realistic commodity-CSI WiFi attains 0.098 m ATE against 0.102 m for LiDAR-A, i.e. a statistical tie, and it is comfortably better than LiDAR-B (0.483 m). This is the paper's central positive result, and it should be stated carefully: WiFi does not *beat* LiDAR at localization, it *matches* it — while costing 84–600× less (Section VI). Parity at one percent of the cost is the interesting claim; a victory would have been a stronger one, and we do not make it.

The parity is also scene-dependent. In the dense street canyon, where roughly nineteen multipath components must be resolved by a four-antenna array, WiFi degrades to 0.314 m while LiDAR-A holds 0.026 m. WiFi's localization is therefore decimetre-class in general and LiDAR-class in sparse multipath, and the honest summary is that it is *sufficient*, not superior.

LiDAR wins mapping, and WiFi does not merely lose — it scores zero. LiDAR reaches occupancy IoU up to 1.00; realistic WiFi reaches ≈ 0 on both scenes. Under oracle sensing WiFi does map (IoU 0.791 on the controlled wall), which localizes the failure squarely in the front-end rather than in the geometry — a thread we pull in Section VIII.

VI. Cost

The replacement question is ultimately economic, so we price it. All prices are dated ranges with sources recorded in the released artifact; we report ranges, never point estimates.

TABLE I

Accuracy on both scenes. WiFi "oracle" uses the ray tracer's true path parameters; "realistic" uses commodity CSI through joint 2-D MUSIC. WiFi rows are *this paper's own runs*, reported as mean $\pm$ std over five seeds; LiDAR rows are single deterministic runs. Lower is better except IoU.

Scene	Sensor	ATE	RPE	Chamfer	m-acc	m-cpl	IoU
Controlled	WiFi oracle	0.049±0.027	0.008	0.505	0.248	0.761	0.791
	WiFi realistic	0.098±0.028	0.012	5.974	2.490	9.458	0.000
	LiDAR-A	0.102	0.030	0.209	0.250	0.168	0.977
	LiDAR-B	0.483	0.055	0.187	0.251	0.123	**1.000**
Street	WiFi oracle	0.104±0.041	0.007	12.344	0.309	24.379	0.077
	WiFi realistic	0.314±0.051	0.014	18.086	3.525	32.647	0.001
	LiDAR-A	**0.026**	0.017	8.674	0.251	17.097	0.163
	LiDAR-B	0.857	0.117	3.734	2.125	5.344	**0.261**

The packages. The vehicle-side WiFi sensing package — a commodity CSI receiver [13] plus antennas — is \$40–95; an ESP32-class node [14] is \$10–35. Ambient access points are treated as pre-existing infrastructure, which is the premise of passive sensing; we test that premise below. LiDAR spans four orders of magnitude, from a legacy spinning unit (\$75,000–80,000) through the mid-range spinning class we simulate (\$8,000–24,000) down to emerging automotive solid-state (\$100–600) and a budget 2-D scanner [15] (\$99). The last of these is a *price floor*, not a peer: it is a 2-D indoor device, and we never compare mapping quality against it.

The gap. Against the LiDAR class we actually simulate, the WiFi package is **84–600**× cheaper (Fig. 3); against legacy spinning, 789–2,000×.

Cost-normalized performance (Fig. 4) is where the comparison becomes sharp, because it forces price to be read together with accuracy. On localization value (price × ATE, lower better) WiFi delivers 4–30 \$·m against LiDAR's 208–2,448 \$·m — one to two orders of magnitude more accuracy per dollar. The gain comes from price, not from precision: WiFi *matches* the LiDAR's accuracy (Section V) while costing two orders of magnitude less. On mapping value (price / IoU) WiFi's cost is *infinite* — it buys no map coverage at any price. LiDAR is the only modality here that converts money into map.

The claim is conditional, and we state the condition. The advantage rests on the ambient-AP premise. If a deployment

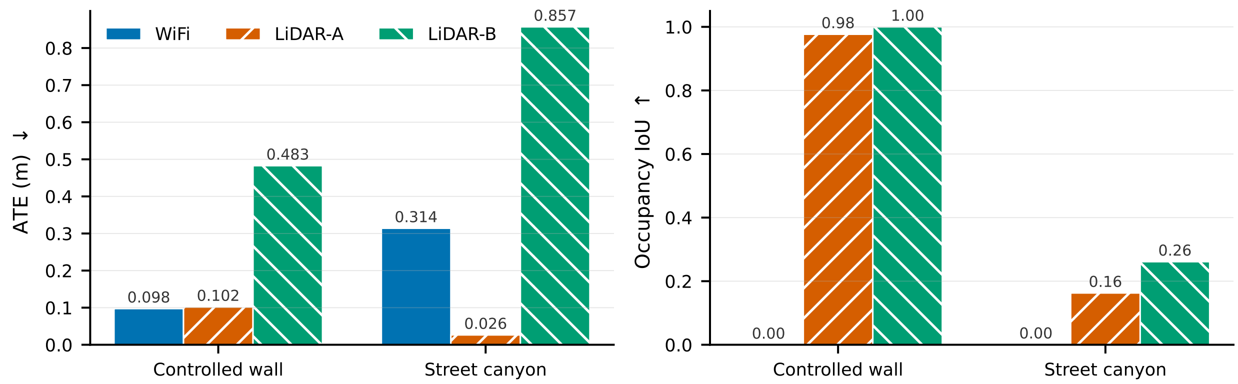

Fig. 2. WiFi wins localization (left) and loses mapping outright (right). Realistic commodity-CSI WiFi beats both LiDAR models on the controlled scene's ATE, yet produces an occupancy IoU of zero on both scenes.

must install its own access points, the WiFi package rises to \$130–335 for three APs and can become *more expensive* (0.3–1.5×) than the cheapest solid-state LiDAR. The dramatic cost story is a story about *infrastructure that already exists*, and it should be quoted that way.

VII. Fusion: Does Keeping Both Pay?

Every prior WiFi+LiDAR system we found demotes WiFi to loop closure while LiDAR maps [5], [4], [9]. Section V inverts that premise — here WiFi is the better *localizer* — so we fuse symmetrically and let the data decide. In the *tight* scheme a single particle filter carries two independent likelihoods, the WiFi bistatic reprojection and a LiDAR scan-match, and the output map is the union of both sensors' points. A deliberately naive *loose* baseline (equal-weight trajectory average, map union) tests whether the tight coupling earns its complexity.

Fusion helps, conditionally (Table II, Fig. 5). In three of four configurations tight fusion beats *both* solo sensors — genuine reinforcement, not mere addition. Against LiDAR-B it improves ATE by 79% on both scenes.

And it can hurt. In the fourth configuration (street, LiDAR-A) the LiDAR alone achieves 0.027 m and fusion degrades it to 0.218 m — an 8× regression. The cause is structural, not incidental: with a fixed equal-confidence weighting, a WiFi likelihood that is ten times less accurate than the LiDAR's pulls the filter off a good solution. We report this rather than tuning the weighting until it disappears. The honest remedy — weighting each modality by its online-estimated reliability — is future work, not a result we claim.

The condition is sensor parity. Fusion reinforces when the two sensors are of comparable accuracy (or WiFi is better) and harms the stronger sensor when they are badly mismatched.

The map. The union adds coverage exactly where LiDAR is incomplete (street IoU 0.149→0.177 for A, 0.262→0.309 for B) but mildly pollutes an already-good map (controlled 0.977 → 0.913): noisy WiFi reflectors fill gaps, they do not sharpen detail.

Is it worth paying for both? The WiFi package is a 0.2–1.2% addition to a LiDAR of the class we simulate. So when fusion helps, it is *essentially free*: ≈ 0.5% more money for a 36–79% better trajectory. This is the strongest practical recommendation in the paper — and it is a recommendation for a *hybrid*, not for a replacement.

TABLE II
Fusion (ATE, m). Tight fusion beats both solo sensors in three of four configurations, and degrades the stronger sensor in the fourth (street/A).

Scene	LiDAR	WiFi only	LiDAR only	Fused (tight)
Controlled	A	0.081	0.102	**0.065**
Controlled	B	0.081	0.212	**0.044**
Street	A	0.281	**0.027**	0.218
Street	B	0.281	0.844	**0.175**

VIII. The Mapping Ceiling

WiFi's mapping failure is the paper's central puzzle: under *oracle* sensing the same geometry maps to 0.25 m [1], yet under realistic CSI it collapses to IoU ≈ 0. Something in the front-end destroys the map. Locating it precisely is this section's contribution — and it turns out that the previously assumed culprit is the wrong one.

A. A learned discriminator cannot fix it

Our prior work [1] attributed the floor to *path discrimination* — telling a genuine facade reflection from LOS, floor and multi-bounce returns — and showed that this label is learnable from per-path features. The natural next step, never taken there, is to close the loop: put the discriminator *inside* the mapping pipeline and see whether the map improves.

It does not. We ran a ladder of filters over the MUSIC detections — none, a physics heuristic (a bistatic-excess gate), a random forest, and a small neural network — gating which detections enter the map while leaving the particle weights (and hence the good localization) untouched. *Every rung leaves the occupancy IoU at* 0.000 *on both scenes*, and the two learned rungs reject essentially everything, emptying the map. Trained on only the features a 2-D front-end can actually measure, the discriminator's held-out F1 is 0.00–0.45, and 0.00–0.20 across scenes.

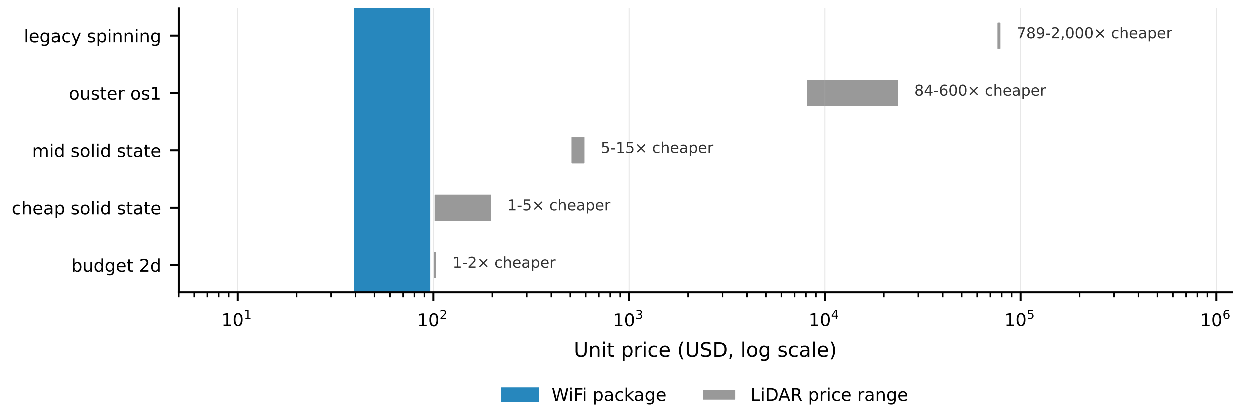

Fig. 3. The sensor-cost envelope (log scale). The WiFi package is 84–600× cheaper than the mid-range spinning LiDAR whose parameters our models simulate. The budget 2-D scanner is shown as a price floor, not as an automotive peer.

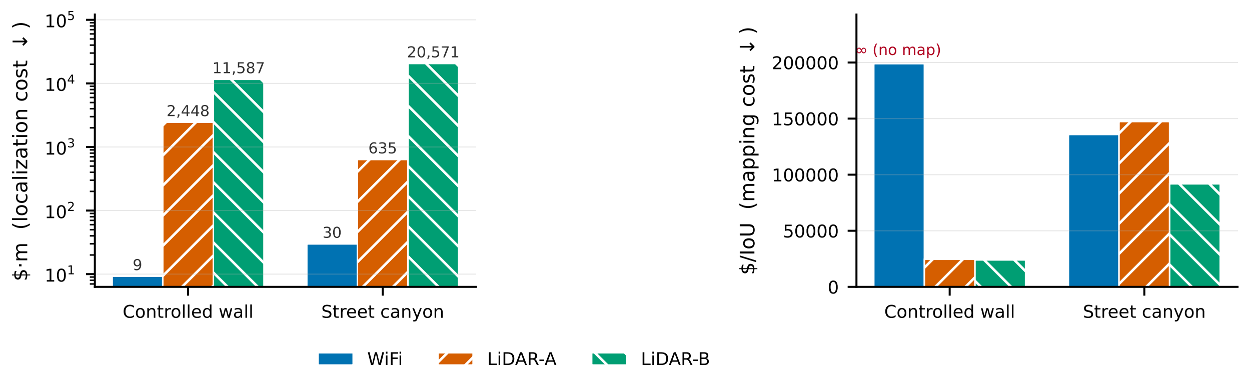

Fig. 4. Cost read together with accuracy. Left: localization cost (price × ATE) — WiFi is one to two orders of magnitude better. Right: mapping cost (price / IoU) — WiFi's is infinite, because it buys no coverage at any price. LiDAR is priced at the tier we simulate.

Two things went wrong in the earlier reasoning. First, the discriminator of [1] used *elevation* as a feature — which a single-ULA delay–azimuth front-end never estimates. It is an oracle quantity, so the reported F1 was optimistic. Second, and more fundamentally, it classified *true ray-traced paths*, whereas the deployed system must classify *MUSIC detections*. Those are not the same objects, as we now show.

B. Isolating the cause

To separate "we keep the wrong paths" from "the paths we keep are badly estimated", we match every MUSIC detection to its nearest true path and then, for those that match a genuine single-bounce facade path, triangulate *the same path twice*: once with the ray tracer's true delay and azimuth, and once with MUSIC's estimates. Any degradation between the two is then *pure estimation error, with discrimination held perfect*. Table III and Fig. 6 give the result.

Three mechanisms emerge, in order of importance.

1) Phantom detections ($\approx 89\%$) **dominate.** Nearly nine in ten MUSIC detections correspond to *no real propagation path at all*: they are estimator artefacts. This is decisive, because *one cannot discriminate among real paths when most detections are not real paths*. Neither the prior work nor our own first analysis identified this term.

TABLE III
ISOLATING THE MAPPING CEILING. DISCRIMINATION — THE PREVIOUSLY ASSUMED CULPRIT — IS THE SMALLEST OF THREE MECHANISMS.

	Controlled	Street
Phantom (matches no real path)	**89.2%**	**89.5%**
Non-facade real path (*discrimination*)	2.2%	8.4%
True facade path	8.6%	2.0%
Triangulated <1 m, *true* params	100%	100%
Triangulated <1 m, *MUSIC* params	**2.4%**	76.7%
MUSIC range error (median)	6.45 m	2.11 m

2) Estimator range bias. On the controlled wall, holding discrimination perfect, swapping true parameters for MUSIC's collapses triangulation from 100% to 2.4% within 1 m. The median range error is 6.45 m.

That WiFi ranging is poor is not itself news — [6] report metre-scale distance errors and abandon range on exactly these grounds. What is new is the *character* of the error. The measured quantity is a *bistatic path length*, whose Rayleigh resolution is c/B (not the monostatic $c/2B$), i.e. 1.87 m at

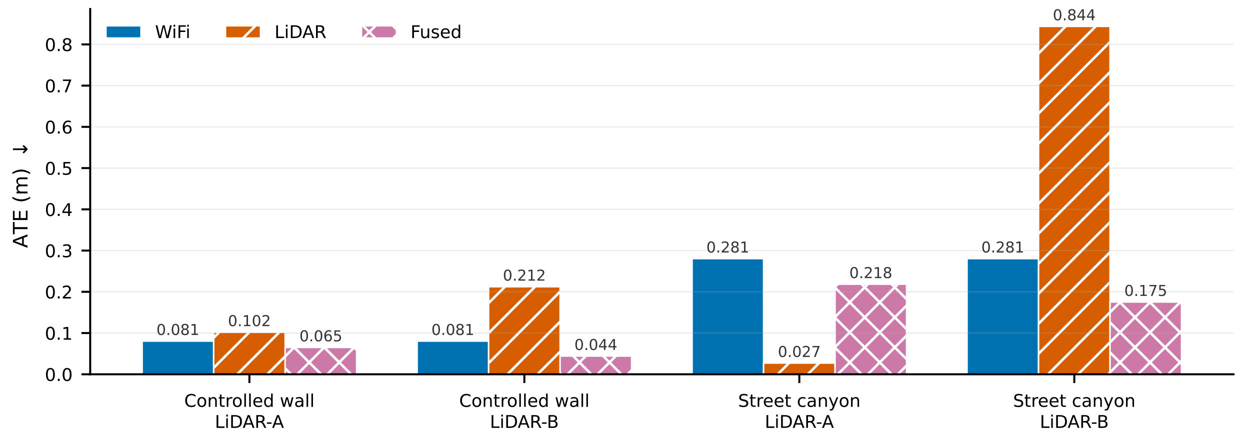

Fig. 5. Fusion is conditional. Tight fusion beats both solo sensors in three of four configurations, but in street/LiDAR-A — where the LiDAR is an order of magnitude more accurate than WiFi — equal-confidence fusion drags the estimate off a good solution.

160 MHz. At 6.45 m, the error is 3.4 times that limit: it is therefore a **bias, not a resolution bound**, and *no amount of bandwidth removes it*. This directly contradicts the natural reading of the bandwidth argument — that a wider channel would fix ranging — and it is corroborated independently: [1] found that 60 GHz with 1.76 GHz of bandwidth (a $44\times$ increase) and sixteen antennas did not move map accuracy at all.

3) Discrimination failures (2–8%)**.** MUSIC does sometimes pick real but non-facade paths. This is the mechanism [1] identified. It is real — and it is the *smallest* of the three.

C. Consequences

A filter *selects* among detections. It can neither *invent* the real paths that 89% of detections fail to correspond to, nor *correct* the bias that ruins the rest. That is why every rung of the ladder failed, and why the failure is not a deficiency of the classifier but of the problem formulation.

This does not show that learning cannot help — only that *classifying estimated paths* cannot. It also says where a learned method must act: at the front-end, suppressing phantoms and correcting bias, or bypassing the estimator entirely to learn CSI $\rightarrow$ geometry end-to-end. It is precisely the latter form that the successful RF-sensing literature takes [12], [11], [10], which in hindsight is not a coincidence.

Finally, a constructive note. On the street, 76.7% of *correctly matched* facade paths do triangulate within 1 m. The geometry is therefore recoverable; the ceiling is set by the front-end, not by the physics of the channel.

IX. Discussion and Limitations

This is a simulation study. Every WiFi and LiDAR result is ray-traced. The one real-data element is the KITTI anchor of Section IV, which validates the SLAM back-end but not the channel model. A ray tracer is a model of propagation, not propagation; the phantom-detection rate in particular is a property of MUSIC operating on *simulated* CSI, and confirming it on real vehicular CSI is the single most valuable next measurement.

The comparison plane is 2-D. Reducing LiDAR to a horizontal ring is what makes the comparison apples-to-apples — both sensors, one plane, one ground truth — but it also denies LiDAR its third dimension. A 3-D LiDAR would map strictly better than the models here; since LiDAR already wins mapping decisively, this understates rather than overstates its advantage, and does not threaten our conclusion.

The cost claim rests on the ambient-AP premise. It is an assumption about the world, not a measurement, and Section VI shows the conclusion inverts without it.

The LiDAR envelope brackets; it does not reproduce a specific unit. We price LiDAR at the tier whose parameters we simulate, and deliberately do not credit the accuracy of a mid-range spinning sensor to the price of a budget solid-state one.

Fusion is not yet safe to deploy naively. The equal-confidence weighting that helps in three configurations harms in the fourth.

X. Future Work

The results point at three concrete next steps. *At the front-end*, where the ceiling actually sits: suppress phantom detections and correct the range bias — a learned *regressor* on the estimator's output, or an end-to-end CSI $\rightarrow$ geometry model of the kind already successful in RF sensing [12], rather than a classifier over estimated paths. *In fusion*: confidence-adaptive weighting, so that a weak sensor cannot drag down a strong one. *In validation*: real on-vehicle CSI, to test whether the phantom rate survives contact with a real channel.

XI. Conclusion

Ambient WiFi can replace LiDAR for one half of SLAM and not the other, and the boundary is sharp. For *localization* it is a genuine drop-in: it *matches* LiDAR in sparse multipath (0.098 m against 0.102 m) and is decimetre-class in general, at 84–600$\times$ lower sensor cost — one to two orders of magnitude better accuracy per dollar. Parity at one percent of the price, not superiority, is the claim; it is the cost ratio that makes

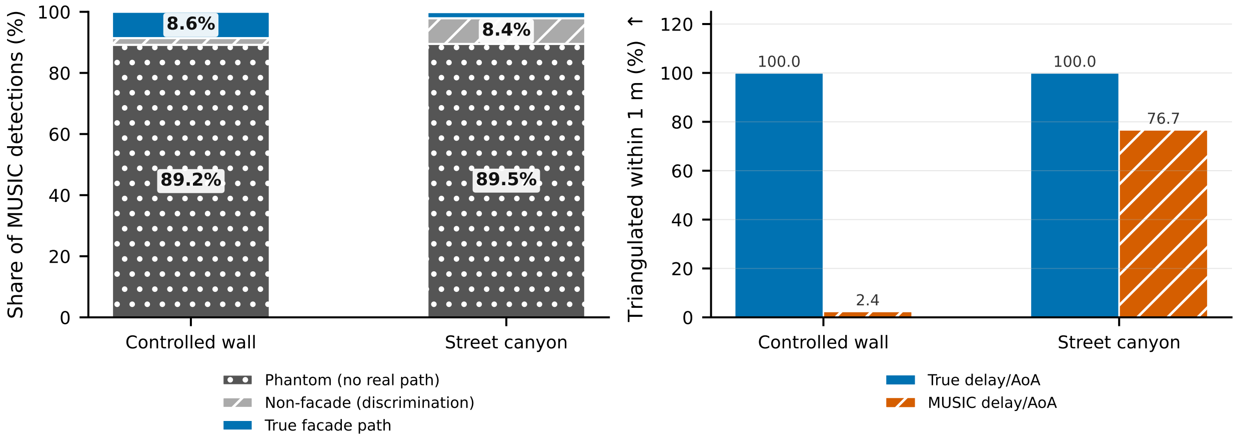

Fig. 6. The mapping ceiling. Left: where MUSIC detections come from — $\approx 89\%$ correspond to no real propagation path. Right: the same true facade paths, triangulated with true versus MUSIC parameters; on the controlled wall a 6.45 m median range bias collapses success from 100% to 2.4%.

it consequential. For *mapping* WiFi fails completely, and no price buys coverage it cannot produce.

That failure is not, as previously supposed, a matter of choosing the right propagation paths. It is a front-end limit: roughly 89% of commodity-CSI detections correspond to no real path at all, and a range *bias* well beyond the resolution limit corrupts most of the rest. A learned discriminator therefore cannot lift the map — a heuristic, a random forest and a neural network fail identically — and any method that hopes to must act on the estimator, not on its output.

The practical recommendation that follows is a hybrid rather than a replacement: WiFi is a 0.5% addition to the cost of a LiDAR that improves its localization by 36–79%, so long as the two sensors are of comparable accuracy. The cheapest useful thing to do with ambient WiFi is not to remove the LiDAR — it is to help it.

Reproducibility

All code, configurations, result artifacts, and the scripts that generate every figure in this paper are released openly. Each quantitative claim is reproduced from a committed data file; the mapping between paper sections and artifacts is tabulated in the repository.

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